

CALINOR ORAL POWDER

Composition:

Each sachet contains:
Calcium Polystyrene Sulfonate 5g

Product Description:

Appearance of powder:
A cream to light brown, fine powder.

Appearance of solution:
Cream to light brown suspensions with insoluble powder.

Pharmacodynamics:

CALINOR is a cation exchange resin which releases calcium in the intestines and binds potassium. This reduces the absorption and metabolic availability of potassium. Since CALINOR is calcium-typed resin, it is used even for patients who restricted ingestion of sodium. Moreover, it can be used without fear of appearance and aggravation of edema, hypertension or cardiac insufficiency induced by sodium patients. Unlike sodium-typed resin, CALINOR does not cause the increase in serum sodium and phosphate levels and the decrease in serum calcium level when it is used for the renal failure patients.

Pharmacokinetics:

After administration of CALINOR via oral route, calcium ion of CALINOR is exchanged for potassium ion in the intestinal tract, particularly around the colon, and the drug is excreted as unchanged polystyrene sulfonate resin into the feces without digestion and absorption. In consequence, potassium in the intestinal tracts excreted outside the body.

Indications:

Hyperkalaemia resulting from acute or chronic renal failure.

Recommended dosage:

CALINOR is for adult use only.

Oral administration

For adults, the daily dosage is 15 - 30 g to be taken dividing into 2 - 3 times. Each dose should be suspended in 30 - 50 mL of water and administered orally. The dosage may be adjusted according to the patient's condition.

Administer calcium polystyrene sulfonate at least 3 hours before or 3 hours after other oral medications. Patients with gastroparesis may require a 6-hour separation.

Route of Administration:

Oral

Contraindication:

- In patients with plasma potassium levels below 5mmol/litre.

- Conditions associated with hypercalcaemia (e.g. hyperparathyroidism, multiple myeloma, sarcoidosis or metastatic carcinoma).
- History of hypersensitivity to polystyrene sulfonate resins.
- Obstructive bowel disease.
- CALINOR should not be administered orally to neonates and is contraindicated in neonates with reduced gut motility (post-operatively or drug-induced).
- Hypersensitivity to the active substance or to any of the excipients.

Warning and Precaution

Binding to other orally administered medications: CALINOR may bind to orally administered medications, which could decrease their gastrointestinal absorption and efficacy. Avoid co-administration of CALINOR with other orally administered medications. Administer CALINOR at least 3 hours before or 3 hours after other oral medications. For patients with gastroparesis, a 6-hour separation should be considered.

Sorbitol: Gastrointestinal stenosis, intestinal ischemia and its complications (necrosis and perforation) may occur in patients treated with polystyrene sulfonate, especially in patients using sorbitol. Therefore concomitant use of Sorbitol with calcium polystyrene sulfonate is not recommended.

Hypokalaemia: The possibility of severe potassium depletion should be considered and adequate clinical and biochemical control is essential during treatment, especially in patients on digitalis. Administration of the resin should be stopped when the serum potassium falls to 5mmol/litre.

Other electrolyte disturbances: Like all cation-exchange resins, calcium polystyrene sulfonate is not totally selective for potassium. Hypomagnesaemia and/or hypercalcaemia may occur. Accordingly, patients should be monitored for all applicable electrolyte disturbances. Serum calcium levels should be estimated at weekly intervals to detect the early development of hypercalcaemia, and the dose of resin adjusted to levels at which hypercalcaemia and hypokalaemia are prevented.

Other risks: In the event of clinically significant constipation, treatment should be discontinued until normal bowel movement has resumed. Magnesium-containing laxatives should not be used.

The patient should be positioned carefully when ingesting the resin, to avoid aspiration, which may lead to bronchopulmonary complications.

Children and neonates: In neonates, calcium polystyrene sulfonate should not be given by the oral route. In children and neonates, particular care is needed with rectal administration as excessive dosage or inadequate dilution could result in impaction of the resin. Due to the risk of digestive haemorrhage or colonic necrosis, particular care should be observed in premature infants or low birth weight infants.

Effects and ability to drive and use machine

There are no specific warnings.

Pregnancy and Lactation

No data are available regarding the use of polystyrene sulfonate resins in pregnancy and lactation. The administration of CALINOR in pregnancy and during breastfeeding therefore is not advised unless, in the opinion of the physician, the potential benefits outweigh any potential risks

Drug Interaction

Orally administered medications: CALINOR has the potential to bind to other orally administered medications. Binding of CALINOR to other oral medications could cause decrease in their gastrointestinal absorption and efficacy. Dosing separation of CALINOR from other orally administered medications is recommended.

Concomitant use not recommended.

Sorbitol (oral or rectal): Concomitant use of Sorbitol with calcium polystyrene sulfonate is not recommended due to cases of intestinal necrosis and other serious gastrointestinal adverse reactions, which may be fatal

To be used with caution

Cation-donating agents: may reduce the potassium binding effectiveness of CALINOR.

Non-absorbable cation-donating antacids and laxatives: There have been reports of systemic alkalosis following concurrent administration of cation-exchange resins and non-absorbable cation-donating antacids and laxatives such as magnesium hydroxide and aluminium carbonate.

Aluminium hydroxide: Intestinal obstruction due to concretions of aluminium hydroxide has been reported when aluminium hydroxide has been combined with the resin (sodium form).

Digitalis-like drugs: The toxic effects of digitalis on the heart, especially various ventricular arrhythmias and A-V nodal dissociation, are likely to be exaggerated if hypokalaemia and/or hypercalcaemia are allowed to develop (see section Special warnings and precautions for use).

Lithium: Possible decrease of lithium absorption.

Levothyroxine: Possible decrease of levothyroxine absorption.

Adverse effect

Metabolism and nutrition disorders: In accordance with its pharmacological actions, the resin may give rise to hypokalaemia and hypercalcaemia, and their related clinical manifestations. Cases of hypomagnesaemia have been reported. Hypercalcaemia has been reported in well dialysed patients receiving calcium resin, and occasionally in patients with chronic renal failure. Many patients in chronic renal failure have low serum calcium and high serum phosphate, but some, who cannot be screened out beforehand, show a sudden rise in serum calcium to high levels after therapy with calcium resin. The risk emphasises the need for adequate biochemical control.

Gastrointestinal disorders: Gastric irritation, anorexia, nausea, vomiting, constipation and occasionally diarrhoea may occur. Gastrointestinal concretions (bezoars) following oral administration have been reported. Gastrointestinal stenosis and intestinal obstruction have also been reported, possibly, due to coexisting pathology or inadequate dilution of the resin. Gastrointestinal ischemia, ischemic colitis, gastro-intestinal tract ulceration or necrosis,

which could lead to intestinal perforation have been reported which is sometimes fatal. The majority of cases have been reported with concomitant use of Sorbitol.

Respiratory, thoracic and mediastinal disorders: Some cases of acute bronchitis and/or broncho-pneumonia associated with inhalation of particles of calcium polystyrene sulfonate have been described.

Overdose and Treatment

Biochemical disturbances resulting from overdosage may give rise to clinical signs and symptoms of hypokalemia, including: irritability, confusion, delayed thought processes, muscle weakness, hyporeflexia, which may progress to frank paralysis and/or apnea. Tetany may occur. Electrocardiographic changes may be consistent with hypokalemia or hypocalcemia; cardiac arrhythmias may occur. Appropriate measures should be taken to correct serum electrolytes (potassium, calcium, magnesium), and the resin should be removed from the alimentary tract by appropriate use of laxatives or enemas.

Storage Condition

Store below 30°C

Storage condition after reconstitution: Use immediately after reconstitution with water.

PRESENTATION

Powder for solution, 5g per sachet. 10, 26, 30, 100 and 126 sachets per pack

Manufacturer and Product Registration Holder

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